Enhancing Early Detection of Infective (Bacterial) Endocarditis in ICU Patients Using Machine Learning: Insights from the MIMIC-III Database

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Introduction

- Infective Endocarditis (IE): Life-threatening infection of heart valves with high morbidity and mortality (Baddour et al., 2015) . Early diagnosis is critical to improve outcomes.
- Diagnostic Challenge: Initial symptoms often non-specific; traditional Duke criteria rely on clinical/microbiological data but can miss early cases (Naderi et al., 2018)..
- High Misdiagnosis Rate: >54% of IE cases are initially misclassified by clinicians (Newman-Toker et al., 2019), leading to delayed treatment.
- Study Aim: Leverage machine learning models to enhance early IE detection using patient data (MIMIC-II ICU database). Evaluate model performance in distinguishing IE vs. non-IE cases.

Study Objective

- Build ML models to classify IE vs. non-IE from administrative/demographic data.
- Use 5 classifiers: Decision Tree, Random Forest, Logistic Regression, XGBoost, CatBoost.
- Compare performance with recent literature.

Literature Review – Marseille Score

- Richet et al. (2008) developed the Marseille score: moderate AUC (~0.69).
- Based on history, physical exam, and basic labs.
- Updated by Edouard et al. (2018) – still moderate accuracy.
- Lacked early-stage sensitivity (Paixão et al., 2024).

Literature Review – EndoPredict-Dx

- Developed by Paixão et al. (2024) for suspected IE admissions.
- Used 10 clinical/lab features (e.g., CRP, hematuria, symptom duration).
- Achieved AUC of 0.78 (95% CI: 0.75–0.81).
- Outperformed Marseille score on same dataset.

Literature Review – SABIER Score

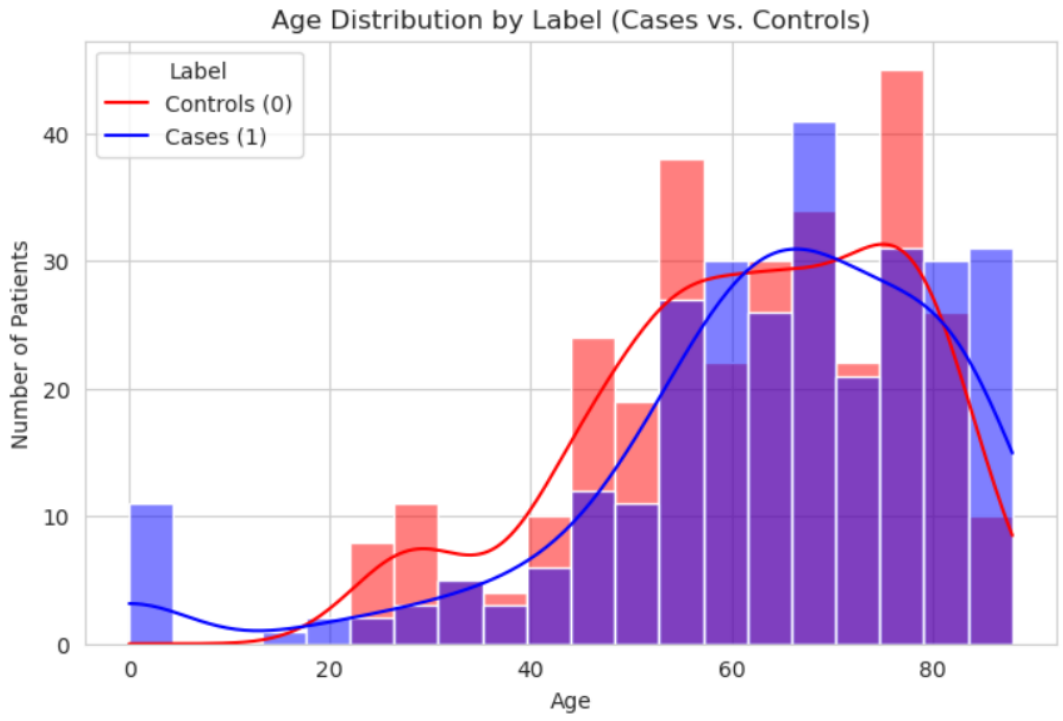
- Lai et al. (2024): ML model for IE in Staphylococcus aureus bacteremia (SAB).
- Trained Random Forest on >15,000 SAB patients; 658 developed IE.
- AUC ~0.74; NPV 98%.
- Focused on ruling out IE in low-risk SAB patients.

Data Preparation

- Data Source: MIMIC-II clinical database (ICU patients), focusing on cases with and without IE diagnosis.
- Dataset: 774 patient records, 30 features per patient (demographics, admission details, diagnoses, outcomes).
- Preprocessing: Removed non-relevant identifiers and timestamp fields to ensure privacy and relevance. Addressed missing values in fields like DEATHTIME, RELIGION, etc., by imputation or exclusion.
- Categorical Encoding: Converted categorical variables to numeric via one-hot encoding, expanding feature count to ~1,211 columns. This prepared the data for machine learning algorithms.
- Outcome Label: Binary classification target – presence of infective endocarditis vs. no IE, determined from discharge diagnosis codes.

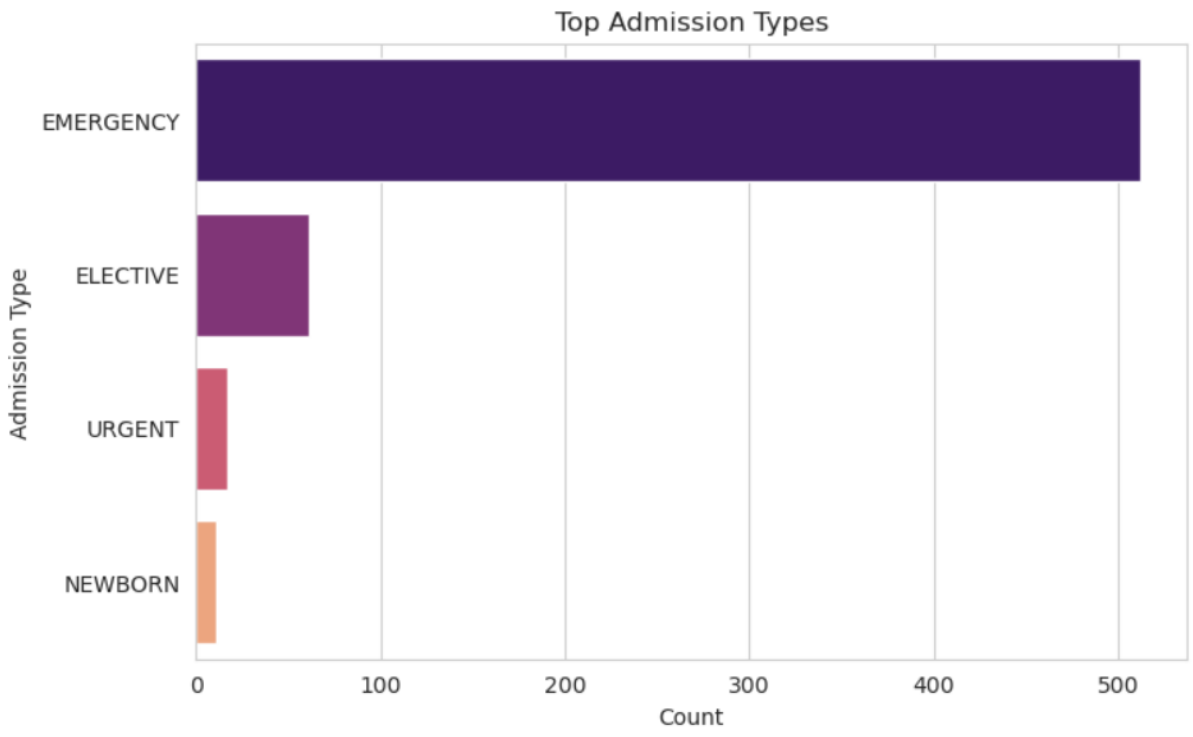
EDA – Age Distribution By Lable

- Overlap in Middle to Older Ages (50–80 years):Both cases and controls are concentrated between ages 50 and 80, with a slight peak around 70–75 years.
- More Young IE Cases (<20 years):There are a few IE cases in younger age groups, where controls are almost absent.
- Controls More Common Around 55–60:Slightly higher number of controls in the 55–60 age range.
- Curve Analysis:
  - The blue KDE curve (cases) suggests IE affects a broader age range.
  - The red KDE curve (controls) is more narrowly peaked, indicating a tighter age cluster.



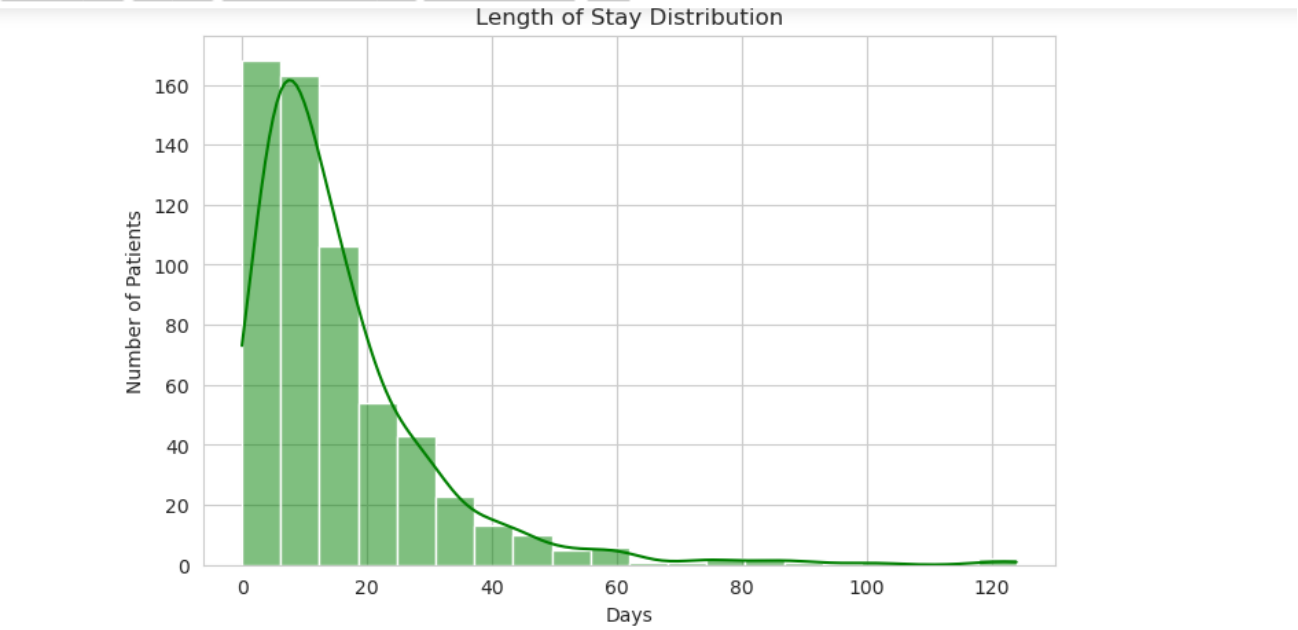
EAD – Top Admission Types

- Emergency Admissions Dominate:
  - Over 500 patients were admitted through the emergency department, making it the most common type by far.
- Fewer Elective and Urgent Cases:
  - Elective admissions were the second most common (~70), followed by urgent admissions (~30), indicating fewer planned entries for the condition studied.
- Minimal Newborn Admissions:
  - Very few newborn admissions were recorded, which is expected given the adult-focused nature of infective endocarditis (IE).



EDA – Length Of Stay

- Right-Skewed Distribution:
  - Most patients had a short stay, with the majority discharged within 10 to 20 days.
- Median LOS is 11 days,
  - while the mean is ~15.7 days, indicating some patients had much longer stays, pulling the average up.
- Long Tails:
  - A few patients had extremely long stays (up to 124 days), possibly due to complications or comorbidities.
- Summary Stats (from output):
  - 25% of patients stayed ≤ 6 days.
  - 50% (median) stayed ≤ 11 days.
  - 75% stayed ≤ 20 days.
  - Maximum recorded stay was 124 days.

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Out[35]: count    601.000000
         mean      15.707155
         std       15.265018
         min        0.000000
         25%        6.000000
         50%       11.000000
         75%       20.000000
         max      124.000000
         Name: LENGTH_OF_STAY, dtype: float64
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Methodology

- ML Models: Trained five classifiers – Decision Tree, Random Forest, Logistic Regression, XGBoost, and CatBoost – to predict IE from patient features.
- Training: Employed stratified 5-fold cross-validation for reliable performance estimation. Ensured balanced representation of IE cases in folds.
- Evaluation Metrics: Assessed performance using Accuracy, Precision (Positive Predictive Value), Recall (Sensitivity), and F1-score for classification quality. Also calculated Specificity, Negative Predictive Value, and Area Under ROC (AUROC) with 95% confidence intervals.
- Model Tuning: Used default parameters with moderate tuning (e.g., CatBoost with 500 iterations, depth=6) to optimize performance without overfitting. Feature importance analysis was conducted for the best model to interpret key predictors.
- Tools: Analysis implemented in Python (scikit-learn, XGBoost, CatBoost libraries). Data split into training and testing sets (80/20) for initial evaluation, combined with cross-validation for final metrics and stability.

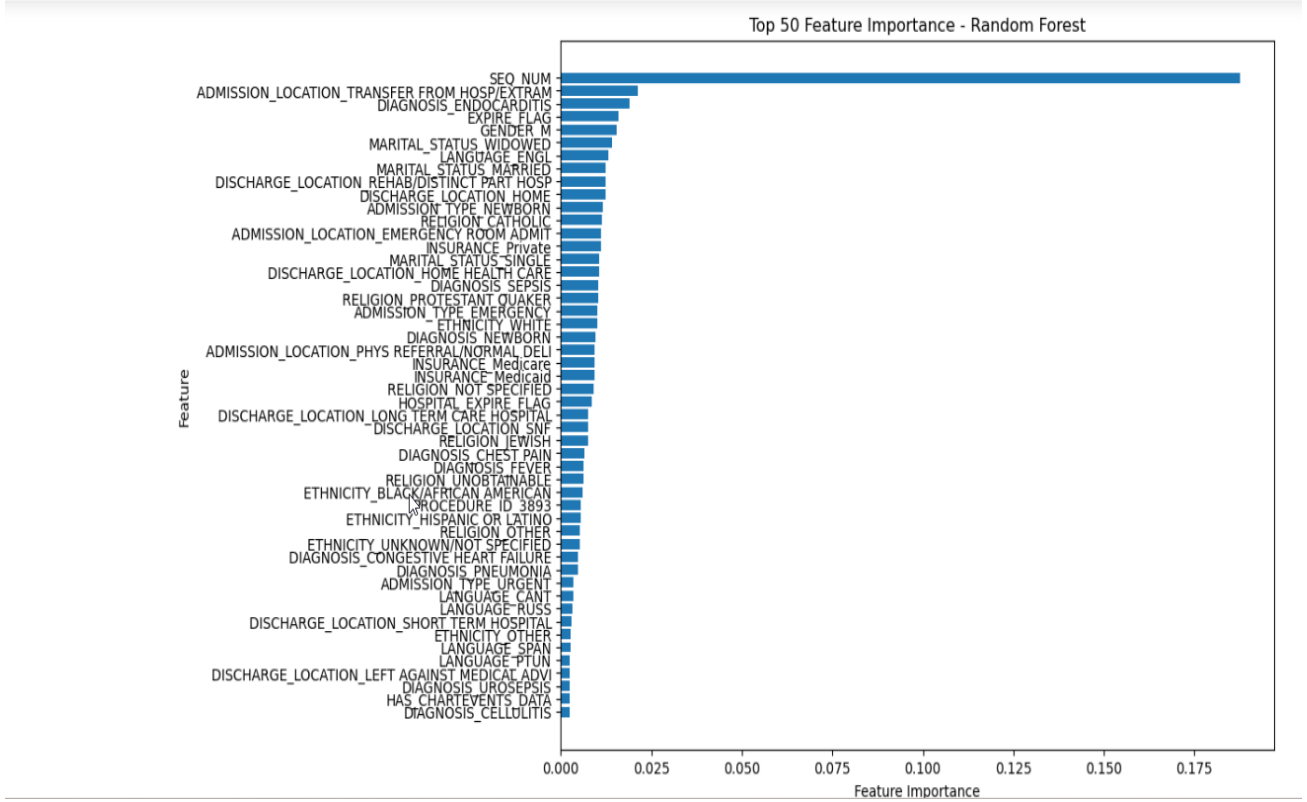
Model Performance

- Random Forest achieved the highest accuracy (84.5%) and had balanced precision and recall (~84.4%), making it the best all-around performer.
- Logistic Regression had the highest recall (89.6%), meaning it caught the most true IE cases, but at the cost of slightly lower precision (~80.2%).
- XGBoost and CatBoost performed identically, both with solid and balanced metrics (~83.9% accuracy, ~83.3% precision/recall).
- Decision Tree had the lowest accuracy (78%) and lowest F1-score (0.79), suggesting it struggled to generalize compared to the ensemble methods.

	Model	Accuracy	Precision (Endocarditis)	Recall (Endocarditis)	F1-Score (Endocarditis)
0	Decision Tree	0.780645	0.759036	0.818182	0.787500
1	Random Forest	0.845161	0.844156	0.844156	0.844156
2	Logistic Regression	0.838710	0.802326	0.896104	0.846626
3	XGBoost	0.838710	0.833333	0.844156	0.838710
4	CatBoost	0.838710	0.833333	0.844156	0.838710

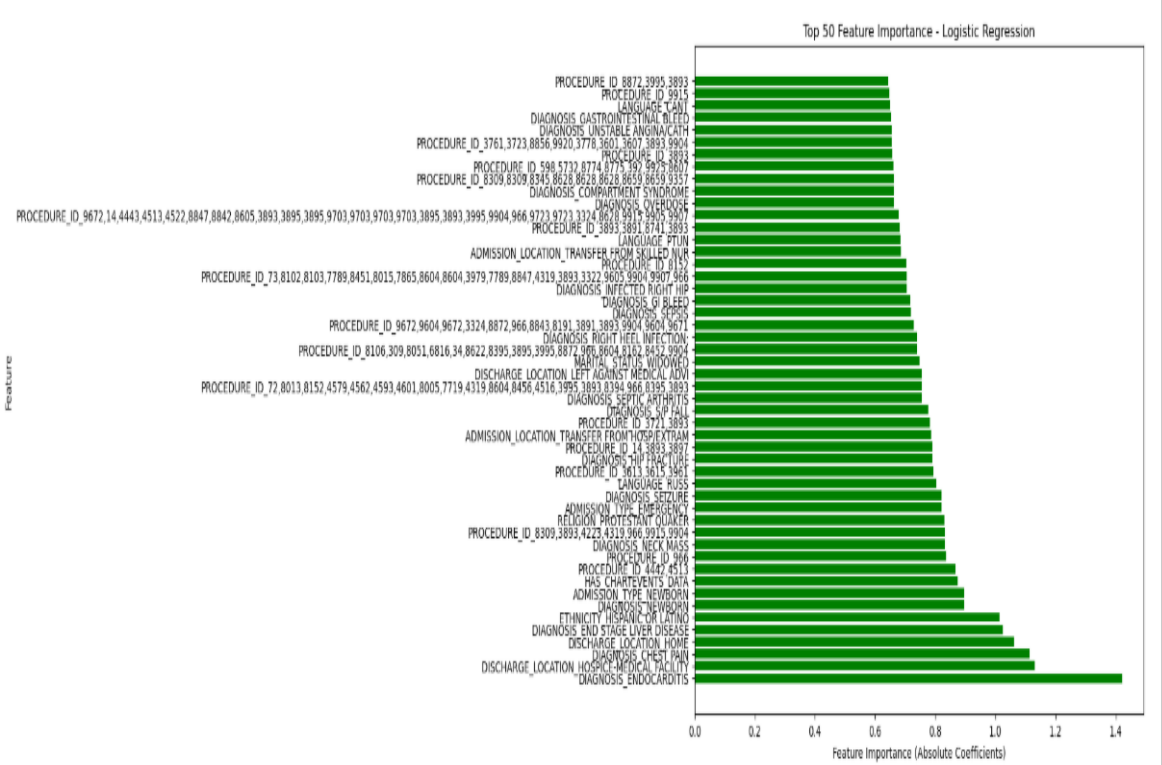
Feature Importance – Random Forest

- Top features:
  - Admission location: Transfer from hospital
  - Diagnosis: Endocarditis
  - Gender, marital status, language, and discharge location
- More emphasis on demographic and admission metadata, not just medical codes.
- Strong performance due to ability to model nonlinear relationships and feature interactions.
- Feature contributions are more evenly distributed compared to CatBoost.



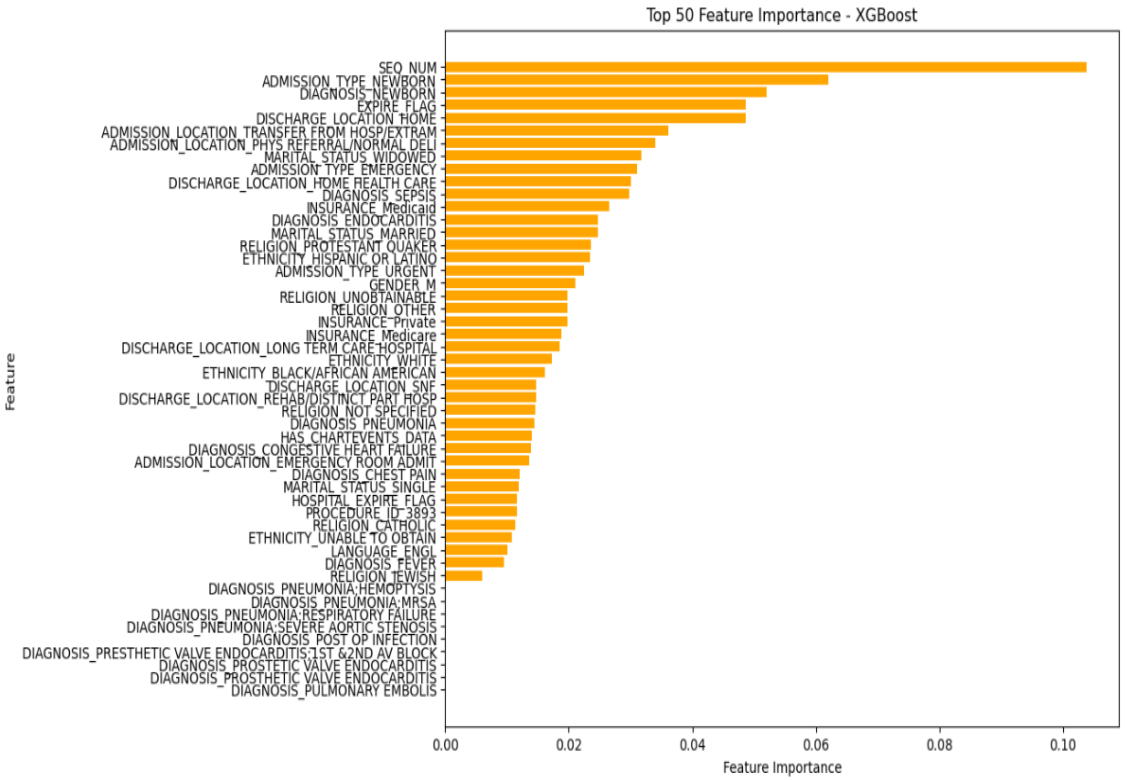
Feature Importance – Logistic Regression

- LR ranks features by the absolute value of their coefficients.
- Top features: multiple procedure codes (e.g., valve surgeries, catheter insertions), admission location, and diagnosis-related features.
- Indicates reliance on structured coded data (procedures, diagnosis descriptions).
- High interpretability, though feature importance is linear and doesn't capture interactions.



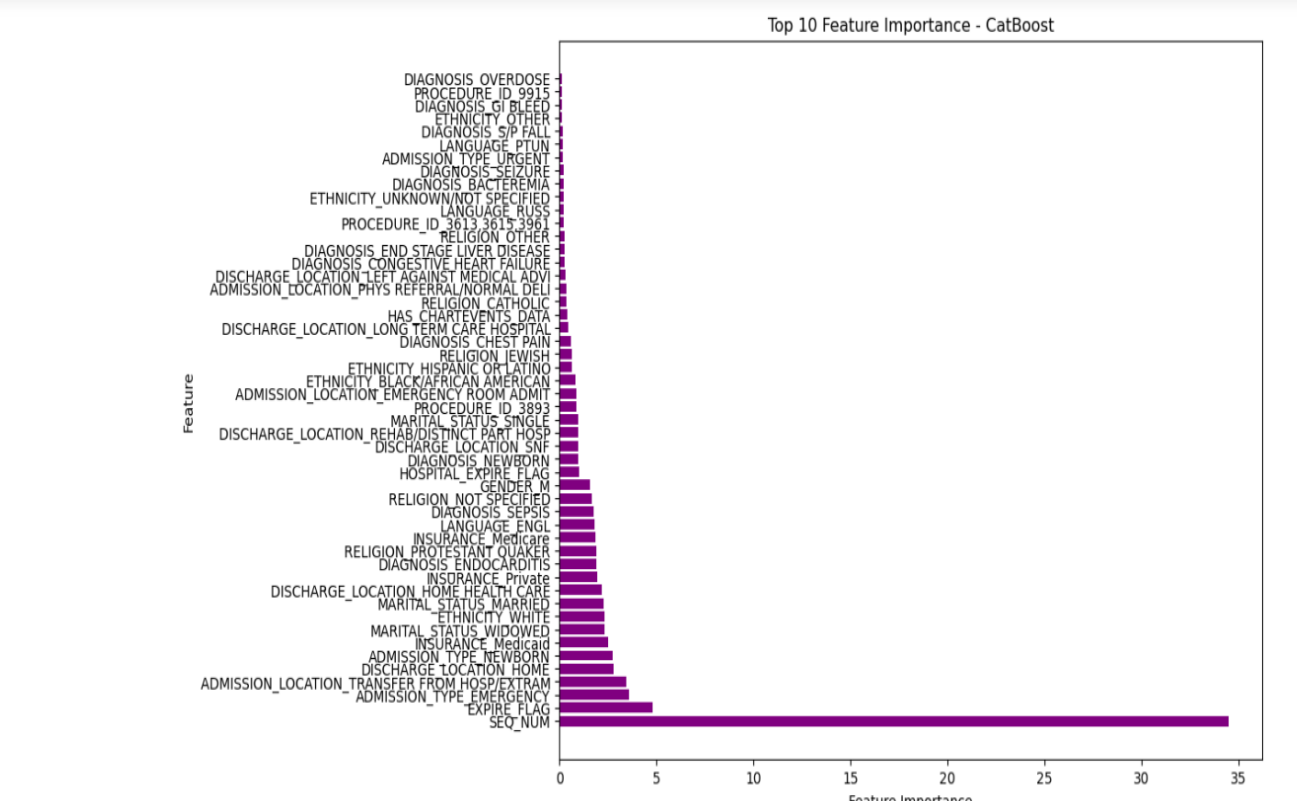
Feature Importance – XGBoost

- Top features:
  - SEQ_NUM
  - Admission type: Newborn,
  - Diagnosis: Endocarditis,
  - Transfer location,
  - Discharge destination
- Similar patterns to Random Forest but with sharper differentiation of importance.
- Emphasizes procedural volume, diagnosis codes, and transition of care, reflecting real-world admission dynamics.



Feature Importance – CatBoost

- Top features by far:
  - SEQ_NUM, EXPIRATION FLAG, Admission Type, Admission Location
- Very steep drop-off after top 2–3 features, suggesting a few features dominate prediction.
- Unique among models in placing very high weight on record sequencing and outcome flags.
- Potential for overfitting if those top features are proxies for discharge-determined labels (e.g., SEQ_NUM or EXPIRE_FLAG might leak post-admission knowledge).



Module Evaluation

- Best Overall AUC: Random Forest, Logistic Regression, and CatBoost all achieved AUC = 0.89, indicating strong discriminative ability.
- Logistic Regression had the highest sensitivity (0.85) and NPV (0.84), suggesting it's best at catching true IE cases and minimizing false negatives.
- CatBoost and XGBoost showed highest specificity (0.81–0.82) and PPV (0.82), indicating they're slightly better at avoiding false positives.
- Decision Tree had the lowest AUC (0.79) and the widest CI (0.74–0.84), meaning its performance is less stable and less accurate.

	Model	AUC	Sensitivity	Specificity	PPV	NPV	95% CI Lower	95% CI Upper
0	Decision Tree	0.79	0.80	0.79	0.79	0.79	0.74	0.84
1	Random Forest	0.89	0.84	0.78	0.79	0.83	0.85	0.92
2	Logistic Regression	0.89	0.85	0.78	0.80	0.84	0.86	0.91
3	XGBoost	0.88	0.80	0.82	0.82	0.81	0.86	0.92
4	CatBoost	0.89	0.83	0.81	0.82	0.83	0.86	0.93

Discussion

- Improved Early Detection: Machine learning models achieved high accuracy and AUROC (~0.89), demonstrating strong potential to support earlier identification of infective endocarditis (IE) and reduce diagnostic delays.
- Comparison with Clinical Standards: ML models outperformed traditional clinical tools like the Duke criteria in early detection, highlighting their value as complementary decision support tools.
- Random Forest vs. Literature: Random Forest stood out with ~85% accuracy, significantly better than the >50% misdiagnosis rate reported in clinical studies (e.g., Naderi et al., 2018), confirming its utility in medical prediction.
- Key Predictors: Features such as emergency admissions, transfers from other hospitals, and diagnoses like sepsis or endocarditis were among the most important, consistent with known clinical risk factors.
- Limitations: The study was limited by its single-center ICU dataset, small sample size, and retrospective design. Broader validation across diverse and larger populations is needed for generalizability.

Conclusion

- Summary: This comparative analysis showed that machine learning models can detect infective endocarditis with high accuracy and reliability. In particular, the Random Forest classifier emerged as the top performer, correctly identifying ~85% of IE cases and achieving an AUC of ~0.89
- Clinical Impact: Implementing such models as decision support tools could significantly improve early IE detection, reducing the high misdiagnosis rates reported in traditional practice and enabling timely treatment. An ML-based alert alongside clinical criteria (e.g., Duke) may prompt further investigation (e.g., echocardiography, blood cultures) in suspected cases sooner.
- Research Contributions: The study provides a comparative benchmark of several ML algorithms on IE prediction. It demonstrates that advanced ensemble methods (XGBoost, CatBoost) and even logistic regression perform on par in this scenario, suggesting that data quality and features play a crucial role.
- Future Work: To move toward clinical adoption, future studies should include larger multi-center data and prospective validation. Integration of additional data (e.g., echocardiographic findings, laboratory values) and real-time model deployment will be important. Ultimately, this approach could be expanded to flag high-risk patients for IE, supporting clinicians in making earlier and more accurate diagnoses.

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